

The Tube

Topological consequences of heterotrophy in the mortal scientist framework

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Abstract

A heterotroph eats in order to reach the token supply held inside another agent. Time spent *exclusively* on extracting that supply is time not spent foraging, mating, or forming and testing hypotheses, and so carries an opportunity cost that selection can be expected to attack. The attack available is concurrency: split the harvest into a fast acquisition phase and a slow internal phase, and let the second run in parallel with everything else the agent does. The internal phase leaves a residue — the part that was not energy — and the residue must leave. Material in, refuse out. This note argues that the heterotroph is therefore, topologically, a *tube*, that the autotroph has no corresponding pressure, and that this is exactly the arrangement observed in the physical world: animals are tubes, plants are not. The argument is set inside the mortal scientist framework, where the pieces it requires — trophic access profiles, interposed media, porosity and partial namespace closure — are already present. Nothing new is assumed. The claim is that the tube is derivable, and in particular that it is a second corollary of the same proposition that already yields “intelligence is a heterotroph’s expense”. The note is a research note in the strict sense: it records an argument, a list of statements to attempt, and the empirical checks each would face.

1 The observation

The reason for eating is to gain access to the token supply held inside the agent being consumed.

There is a distinct evolutionary advantage to minimising the time spent *exclusively* on energy extraction. If some portion of the extraction can be carried on internally to the heterotroph — so that the heterotroph can meanwhile mate, seek further food, form hypotheses about its environment, make predictions — then extraction runs concurrently with everything else the agent does, and that concurrency is worth paying for.

Hence eating has two phases:

1. the gross phase: getting the relevant material across the boundary and inside;
2. the internal phase: the remaining processes working on that material while the agent is engaged elsewhere.

The residue of the internal phase is the part that was not energy. It must leave. Material in, refuse out: this makes the heterotroph, topologically, a tube.

The autotroph has no such need. Nothing about harvesting an external source calls for a tube structure. And this is the arrangement found in the physical world.

2 What the framework already supplies

Every piece the argument needs is on the table. The claim of this note is not that a new mechanism is required but that the tube *follows*.

Access profiles. The scientist note’s trophic section distinguishes a source — a persistent emitter, rate-limited, non-exclusive, non-adaptive — from prey — a linear send, exhaustible, exclusive, adaptive. Same sort, same rewrite rule, different multiplicity. Trophic type is a fact about the cut, not a new kind of thing.

The asymmetry that generates the refuse. A source emits *tokens*. Prey is a *computation*: a structured term that happens to hold a stack. Harvesting a source yields token and nothing else; breaking prey open yields token entangled with code.

Observation 1 (Refuse is not an extra postulate). The residue exists precisely because prey are computations and sources are not. It is the non-token part of a structured prey term. No assumption beyond Table 1 of the scientist note is required to produce it.

Media and interposed channels. The engine note’s error model generalises $\text{for}(y \leftarrow x) P \mid x!(Q)$ to $\text{for}(y \leftarrow x1) P \mid C(x1, x2) \mid x2!(Q)$, and ranks media by behaviour: relay, lossy, corrupting, reading, converting, with the engine at the top rung. A d -hop pathway is already constructed there.

Porosity and individuation. An individual is Chn_{in} closed, $\text{Chn}_{\text{boundary}}$ open. Boundary channels require redex pathways to internal channels or they are stranding sites. Integration depth d costs reliability p^d ; ownership is what buys depth.

The tower. The outside of one level is the inside of the next.

Namespace separation as the guarantor of non-interference. In the game note, threat offers had to live on their own names: a line signalling on a square’s move channel would rendezvous with that square’s receipt and play the move. Perception must not be able to act.

3 The central identification

Observation 2 (The gut is an internalised engine). The digestive apparatus is a converting medium held inside the individual’s boundary namespace: a chain of interposed channels running from an intake channel to an expulsion channel, with the whole chain owned.

The mycorrhizal network was the external instance of the engine profile. The gut is the same profile drawn on the other side of the ownership line. This is not an analogy between two systems; it is one access profile appearing at two positions relative to a boundary.

The observation that makes the topology non-metaphorical is the next one.

Observation 3 (The lumen is outside). The interior of the tube is not inside the individual. It is boundary namespace: owned, but open.

A tube is therefore the construction that surrounds a region of the outside with the inside. That is the tower's generating step, applied for a specific purpose: to acquire boundary surface without acquiring exposure. The biological commonplace that the gut lumen is topologically external is, on this reading, the same fact as the framework's fractality — not a coincidence to be noted but the mechanism itself.

4 Statements to attempt

Numbering is provisional and every item below is a candidate rather than a result.

Proposition 4 (Separation is not itself yielding). *In the harvest of a structured prey term, the rewrite steps that separate token from residue are metered but token-neutral. They cost and do not pay.*

Proposition 5 (The serialisation penalty). *If separation is sequential with the scientist's control loop, the scientist pays its burn rate over the whole separation with no inquiry, no foraging and no mating during it.*

The target here is to write the foraging inequality with a sequential fraction σ and derive the Amdahl-shaped bound on sustainable harvest rate, so that the advantage of concurrency emerges as a strict inequality with an explicit threshold rather than as a preference.

Proposition 6 (Concurrency forces internalisation). *Concurrency as such is trivially available in ρ : it is parallel composition. The content is therefore not that extraction can be concurrent but where the extraction channels live. If separation runs on boundary channels, other agents may rendezvous with partially processed material. Exclusivity of a partially extracted harvest requires the separation chain to be owned.*

Proposition 6 is the ownership-buys-depth corollary specialised to digestion, and it is the step that converts a remark about scheduling into a claim about topology.

Proposition 7 (Two ends, not one). *Intake and expulsion must sit on distinct names. On a single shared channel an offer of refuse and a receipt for intake rendezvous: the organism re-ingests its residue, or the residue blocks the intake. A single-opening organism is therefore not impossible but rate-limited by a mutual exclusion — intake and expulsion must alternate on the shared name.*

This is the perception-must-not-act argument of the game note, one level up. It has an immediate empirical check, and the check is favourable: cnidarians and flatworms possess a gastrovascular cavity with a single opening and feed in discrete alternating bouts, while bilaterians with a through-gut feed continuously. The framework should predict the throughput gap from the exclusion alone.

Proposition 8 (Autotrophs are not tubes). *For source access, both of the tube's ends lose their reason. The profile is non-exclusive, so nothing is gained by fast acquisition; the source is unstructured, so there is no residue to expel.*

The prediction is not merely that a plant needs no tube but that it should instead buy boundary area directly. A leaf and a root system are that purchase.

Corollary 9 (The second heterotroph’s expense). *The scientist note derives “intelligence is a heterotroph’s expense” from the limit-of-inquiry proposition together with structural recursion. The tube should be a second corollary of the same proposition: exclusivity together with structure forces both purchases.*

If Corollary 9 goes through it is the strongest result available here, because it makes a gross anatomical fact a consequence of an access profile. Intelligence and the tube become siblings rather than neighbours.

Proposition 10 (Transient versus persistent tubes). *A food vacuole is a tube built and dissolved per meal; a gut is one maintained. Persistence carries a maintenance cost against the metabolic stack, so there is a threshold harvest rate above which the permanent tube beats the transient one.*

Amoeba below the threshold, annelid above it. The composite-resolution proposition already says the overlap region between strategies is populated by composites rather than atoms, so a graded answer here is expected rather than a disappointment.

Proposition 11 (Gut depth tracks incommensurability). *Tokens held in a foreign colour are not nutritive as such; they must be converted. The number of interposed converting stages should scale with the colour distance between prey namespace and consumer namespace.*

Carnivores eat close relatives and have short guts. Herbivores eat far ones and have long guts, often renting whole foreign namespaces — symbionts — inside the boundary to perform conversions they cannot perform themselves. A rented engine inside the tube is a market inside an individual, and it survives the Coase objection recorded in the engine note precisely because, by Observation 3, the lumen is not the inside.

5 What to write down formally

The two-phase harvest as a term. An exclusive acquisition rendezvous on a boundary channel, in parallel with a persistent digestion chain on internal channels, terminating in a send on the expulsion channel. The existing `Forage` of the scientist note’s worked ecosystem and the `Chain` of `engine.rho` supply most of it.

The tube as a namespace shape. $\text{Chn}_{\text{boundary}}$ factors as $\text{Chn}_{\text{in}} \sqcup \text{Chn}_{\text{out}}$ with a directed redex pathway from in to out and no return path. Then *tube* is a formula in the logic rather than a designation, which is the same move already made for “being a scientist is a depth- ≥ 2 dialogue loop through a both-type channel”.

Whether the absence of a return path is forced. Coprophagy, caecotrophy and rumination are return paths, and they occur exactly where one pass leaves too much unextracted. This is cheap to state as a condition on residual yield and should be stated, since a proposition that forbids observed behaviour is worse than one that predicts it.

The physical topology as a shadow. A through-gut makes the body a genus-one surface. That is real, but it is not the content. The content is the channel topology; the surface is what an embodiment of that topology in three-space looks like. The note should say so explicitly, so that it does not appear to be arguing from anatomy to mathematics.

6 Open questions

Question 1. Is the tube forced or merely favoured? Composite resolution predicts mixotrophs, and predicts that their tubes should be facultative. Is that borne out?

Question 2. Does the surface-area argument connect to the critical radius r^* of the compression-versus-autonomy result? A tube buys boundary area at fixed volume; r^* is a critical radius. There is plausibly one calculation here rather than two.

Question 3. Where does circulation enter? Beyond a certain body size the tube's yield must be distributed, which is a second internal medium with a different profile — relay rather than converting — and a different topology, branching and returning. If the tube follows from the access profile, does the vascular tree follow from the tube together with the allometric constraint?

Question 4. What is the assay analogue? Tasting is an assay run at the intake channel deciding whether to admit; vomiting is a refutation acted upon after admission. The trichotomy and budget-relative refutation should both apply, and the lossy-medium correction owed upstream to the scientist note's assay section may turn out to be the same correction.

Question 5. Does the tube give a new grading dimension, or is it already covered by the vector over typed namespaces? Suspicion: the tube is a fact about k_{chn} , which would be an independent argument for the fifth namespace.

7 Filing

The natural home is the trophic-structure section of the scientist note, since the material is short and is a corollary of what is there. If Propositions 5, 7 and 11 develop enough measurable content — a threshold, a throughput ratio, a scaling law — the note stands on its own in the rho-life series instead.